

ACKNOWLEDGEMENT

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Lastly, this project was inspired by the need to connect Nepali farmers directly with buyers, cutting out unnecessary middlemen and making fresh produce more accessible. I hope Krishik Bazar contributes to improving market access for farmers and convenience for buyers across Nepal.

ABSTRACT

Krishik Bazar is a farmer-to-buyer produce marketplace web application developed to connect Nepali farmers and sellers directly with buyers. The system enables farmers and sellers to list fresh produce such as vegetables, fruits, grains, and dairy, while buyers can browse listings, add items to a cart, check out, and track their orders. Krishik Bazar comprises all the required functionalities to manage users, products, and orders efficiently. Moreover, the system provides robust authentication and authorization using JSON Web Tokens and OTP-based email verification to ensure that only genuine users can list produce or place orders. Additionally, the system supports a sandboxed multi-method checkout flow simulating eSewa, Khalti, and Cash-on-Delivery, laying the groundwork for real payment gateway integration. Krishik Bazar is built as a MERN stack monorepo, with a React single-page application frontend and an Express and MongoDB backend, designed to be scalable, maintainable, and responsive across devices. Krishik Bazar is applicable for farmers, cooperatives, and small sellers looking to reach buyers directly without relying on intermediaries.

Keywords : *Krishik Bazar, MERN Stack, Farmer-to-Buyer Marketplace, E-Commerce, JSON Web Tokens, MongoDB, React, Express, Node.js, Order Management*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ER	Entity-Relationship
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JS	JavaScript
JWT	JSON Web Token
MERN	MongoDB + Express + React + Node
OTP	One-Time Password
REST	Representational State Transfer
SMTP	Simple Mail Transfer Protocol
SPA	Single Page Application
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
WWW	World Wide Web

CHAPTER 1: INTRODUCTION

1.1 Introduction

Agriculture is central to Nepal's economy, with a large share of the population depending on farming for their livelihood. Traditionally, produce moves from farmers to buyers through a chain of middlemen and local wholesale markets, which increases the final price for buyers while reducing the margin that reaches the farmer. This process is often slow, opaque, and difficult for smallholder farmers to navigate.

With the growth of internet access and smartphone usage across Nepal, digital marketplaces present an opportunity to close this gap. Krishik Bazar is one such solution: a web-based farmer-to-buyer marketplace that allows farmers and sellers to list fresh produce directly, and allows buyers to browse, add items to a cart, check out, and track their orders online.

The system is built to be lightweight and accessible, focusing on a clean listing and ordering workflow rather than complex logistics, so that farmers with modest technical experience can list produce and buyers can shop with confidence.

1.2 Problem Statement

Farmers often lack a direct, digital channel to reach buyers outside their immediate area, limiting their market and bargaining power. Buyers, on the other hand, have few reliable options to purchase fresh produce directly from farmers online, and existing general e-commerce platforms are not tailored to the needs of an agricultural marketplace, such as produce categories, seller-managed listings, and simple order tracking.

To address these issues, a dedicated digital marketplace is needed. The proposed "Krishik Bazar" aims to make buying and selling produce more direct, transparent, and convenient by giving farmers and sellers their own listing tools and giving buyers a familiar cart-and-checkout shopping experience.

1.3 Objectives

To design and develop a secure, farmer-to-buyer online marketplace that allows farmers and sellers to list produce and allows buyers to browse, purchase, and track orders through a MERN stack web application.

1.4 Scope and Limitations

Scope

- It supports user registration and OTP-based email verification, ensuring that farmers, sellers, and buyers are who they say they are, and it manages user profiles for both sellers and buyers.
- Sellers can list, update, and remove produce listings while buyers can browse the marketplace, add items to a cart, check out, and view their order history.

Limitations

- Krishik Bazar currently processes payments in a sandbox mode simulating eSewa, Khalti, and Cash-on-Delivery; real payment gateway credentials are not yet wired in, so live transactions are not supported.

1.5 Development Methodology

Krishik Bazar was developed using an Agile, iteration-based methodology. Rather than completing the system in a single fixed sequence, the project was broken into short cycles covering backend API development, frontend integration, and testing, with each cycle building on and refining the previous one.

1.6 Report Organization

- Chapter 1 is the Introduction. It describes the project, the problem it addresses, and the objectives behind Krishik Bazar.
- Chapter 2 is the Background Study and Literature Review. It discusses the context behind Krishik Bazar and related work in digital agricultural marketplaces.
- Chapter 3 is the System Analysis. It lists the requirements for Krishik Bazar, evaluates feasibility, and analyses how the system is organized.
- Chapter 4 is the System Design. It describes the system's data model, forms and interface design, and deployment structure.
- Chapter 5 is the Implementation and Testing. It explains how Krishik Bazar was built, the tools used, and how it was tested.
- Chapter 6 is the Conclusion and Future Recommendations. It summarizes the outcomes and outlines directions for future development.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

The proposed marketplace, named "Krishik Bazar," is a web-based application that facilitates direct produce trade between farmers/sellers and buyers. Traditional produce distribution in Nepal involves manual coordination and multiple intermediaries, resulting in higher prices for buyers and lower returns for farmers. With the advancement of web technologies, the proposed system is expected to resolve these issues by providing a faster, more transparent buying and selling channel.

One of the major concepts applied in the proposed system is authentication and authorization. Krishik Bazar is implemented using JSON Web Tokens, which ensure the security and reliability of user sessions. Passwords are encrypted using the bcrypt hashing algorithm, which prevents unauthorized access to seller and buyer accounts. Additionally, OTP-based email verification through nodemailer confirms user identity during registration and sensitive actions.

The system design is based on a client-server architecture pattern, where the client-side application is built using React and communicates with the server using RESTful APIs provided by the Node.js and Express backend. The client-side application provides an interactive, responsive interface using Tailwind CSS, with reusable components for product cards, cart, and order views.

Another significant concept in the system design is security in web application development, including CORS configuration restricting the API to the trusted frontend origin, environment-based configuration through dotenv, and encrypted transmission of sensitive data such as credentials and tokens. These features ensure that the marketplace is resilient against common web security threats.

In conclusion, "Krishik Bazar" is a system design that incorporates modern MERN stack development practices, secure authentication, and flexible document-based data management to produce an effective farmer-to-buyer marketplace.

2.2 Literature Review

Traditional produce trade in Nepal is primarily carried out through wholesale markets (haat bazaars) and local vendors, where farmers bring their produce to a central market for sale to traders. While this method is well established and trusted, it is time-consuming for farmers and often results in reduced margins due to multiple layers of intermediaries [1].

Recent growth in e-commerce and digital platforms in Nepal, including general marketplaces and food delivery services, has demonstrated growing consumer comfort with ordering goods online, encouraging the development of sector-specific platforms such as agricultural marketplaces [2].

Globally, agri-tech platforms have emerged that connect farmers directly to buyers and retailers, reducing dependence on intermediaries and improving price transparency for both sides of the transaction [3].

Studies on digital marketplaces highlight that secure authentication and verified seller identity are critical to building buyer trust, particularly for perishable goods such as fresh produce, where buyers rely on the platform's credibility rather than physical inspection before purchase [4].

Research also indicates that document-oriented databases such as MongoDB are well suited to marketplace applications with heterogeneous product attributes, offering schema flexibility that simplifies onboarding new product categories without extensive migrations [5].

Additionally, sandboxed or simulated payment flows are commonly used during the early development of e-commerce systems in Nepal, allowing platforms to validate checkout logic before integrating production credentials for services such as eSewa and Khalti [6].

CHAPTER 3: SYSTEM ANALYSIS

3.1 System Analysis

Analysis of the system requires understanding its requirements, feasibility, and intended function. The "Krishik Bazar" system is intended to facilitate an efficient, secure, and convenient method for farmers and sellers to list produce and for buyers to purchase it online.

3.1.1 Requirement Analysis

Requirement analysis identifies what the system should do and the constraints under which it must operate.

Functional Requirement

- Users can register and log in using email and password with JWT-based authentication, verified via an OTP sent to their email.
- Sellers can create, update, and delete produce listings with category, price, and quantity details.
- Buyers can browse and search products, add items to a cart, and check out within a few seconds of interaction.

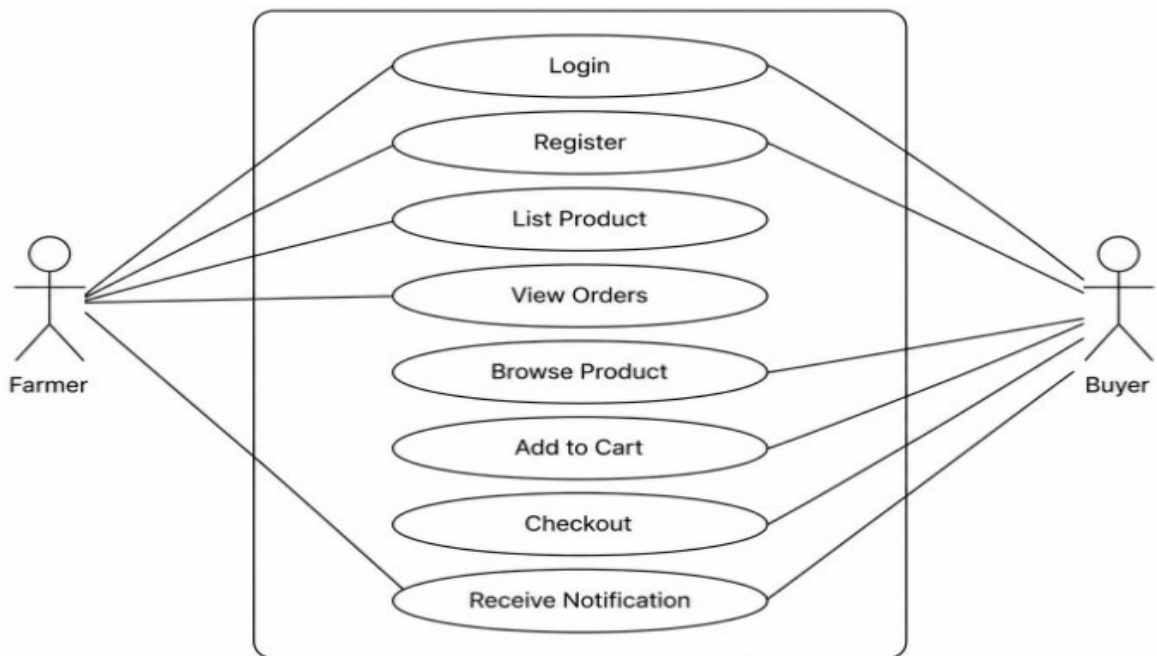


Figure 3.1: Use Case Diagram of Krishik Bazar

Non-Functional Requirement

- Scalability: The system can handle a growing number of users, product listings, and orders without performance degradation, aided by MongoDB's flexible document model.
- Reliability: Ensures accurate order recording and consistent system availability with minimal downtime.
- Security: Implements JWT authentication, bcrypt password hashing, and OTP email verification to protect user accounts and transactions.
- Performance Efficiency: Provides fast response times through indexed MongoDB queries and a Vite-optimized React frontend.

3.1.2 Feasibility Analysis

Feasibility analysis evaluates whether the proposed system can be successfully developed and implemented within available resources and constraints.

Technical Feasibility

The proposed system can be developed using widely available, well-documented technologies. The frontend interface is built with React, Vite, and Tailwind CSS, while the backend is implemented using Node.js and Express. MongoDB, a document-oriented database, is used to store user, product, and order data with flexible schemas suited to varied produce categories.

Operational Feasibility

The system features a simple, familiar e-commerce interface, allowing buyers to browse, add to cart, and check out without needing advanced technical knowledge. Sellers have access to listing management tools that let them add, edit, and remove products and view incoming orders efficiently.

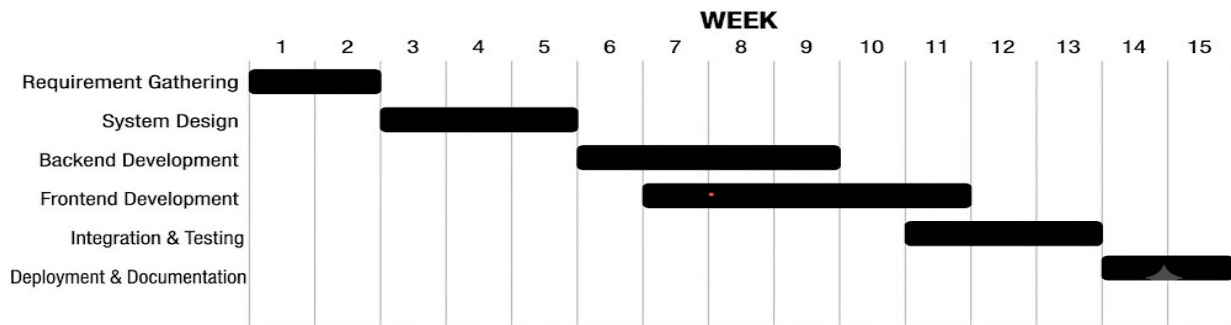
Economic Feasibility

The development of the system primarily requires software tools and development time. Since most of the technologies used, including Node.js, Express, React, and MongoDB, are open-source and freely available, the overall development cost is relatively low.

Schedule Feasibility (Project Timeline)

The project was completed within the academic semester according to the following schedule.

Table 1: Project Schedule



3.2 Analysis

Data Flow Diagram (DFD): At the context level, the primary external entities are the Seller and the Buyer. The Seller sends product listing data to the Krishik Bazar system and receives order notifications in return. The Buyer sends browsing, cart, and checkout requests to the system and receives product listings, order confirmations, and order status updates. Internally, the system processes authentication requests, manages the product catalog, and coordinates order placement and tracking between the Seller and Buyer flows, with all persistent data stored in the MongoDB database.

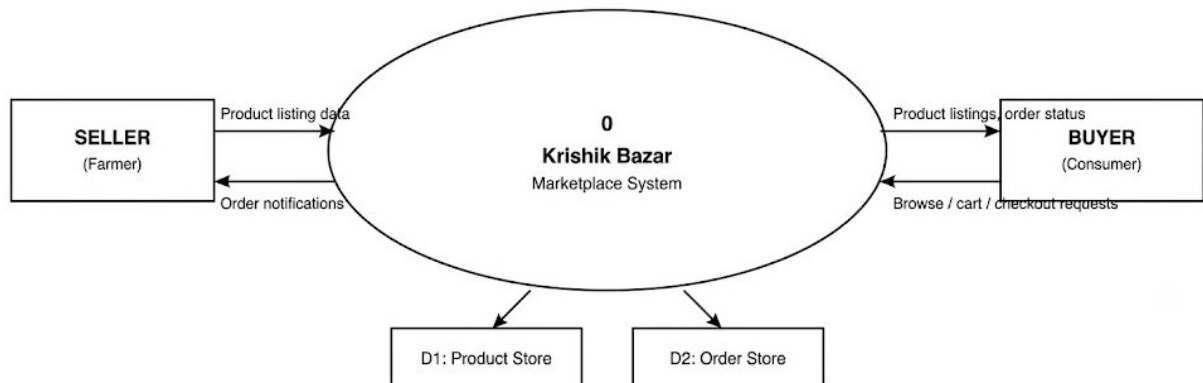


Figure 3.2: Context Diagram of Krishik Bazar

CHAPTER 4: SYSTEM DESIGN

4.1 Forms and Report Design

Forms are used to present or collect information on a single item such as a product, a user, or an order. Forms can be used for both input and output. Reports, on the other hand, are used to convey information on a collection of items, such as a seller's list of active listings or a buyer's order history.

4.2 Interface and Dialogue Design

The primary dialogue that appears when a user first accesses the platform gives them the option between a Login Form and a Register Form, where new sign-ups can register as either a Seller or a Buyer. After authentication, buyers are directed to the Marketplace view, where products are arranged in a responsive grid with category filters, price, and an "Add to Cart" action on each card. The Cart and Checkout flow guides buyers step by step from reviewing cart items, to selecting a payment method (eSewa, Khalti, or Cash-on-Delivery), to order confirmation, and finally to an Order Tracking page showing current status.

4.3 Deployment Diagram

A deployment diagram shows how the software is deployed across hardware or servers. In Krishik Bazar, the React SPA is served from the Vite build (dist/) via a static host or the Vite dev server on port 5173 during development, and communicates over HTTP/REST with the Express API server running on port 8080. The Express server, in turn, connects to a MongoDB instance, either a local mongod process or a MongoDB Atlas cluster, over a Mongoose connection, and connects to an SMTP server (Gmail or another provider) for sending OTP and transactional email. CORS on the backend is configured to accept requests only from the trusted frontend origin, ensuring that the API is not exposed to arbitrary clients.



Figure 4.1: Deployment Diagram of Krishik Bazar

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1 Implementation

The implementation phase focused on building Krishik Bazar based on the structured system design created earlier. Various tools and technologies were utilized to ensure efficient development, testing, and deployment.

Tools Used

CASE Tools

Draw.io: Used for designing UML diagrams such as the ER diagram, use case flows, and the deployment structure. These diagrams helped visualize the system architecture and workflow.

Git & GitHub: Version control and collaboration were managed through Git and GitHub. Git allowed efficient tracking of changes, while GitHub provided a platform for remote collaboration, backup, and code review.

Postman: Used for API testing of the Express backend. Postman ensured that the /auth, /users, /products, and /orders routes responded correctly and securely before frontend integration.

IDEs

Visual Studio Code: Used for both backend and frontend development. Lightweight and highly customizable, VS Code enabled faster development of the Express API and the React interface with Tailwind CSS integration.

Programming Languages

JavaScript: Frontend development used React 19 for building dynamic, responsive, component-driven interfaces, ensuring a smooth shopping experience for buyers. Backend development used Node.js and Express 5 for the REST API and business logic.

Database Platform

MongoDB: All data related to users, products, carts, and orders is stored in MongoDB via Mongoose 9. MongoDB is a widely used, open-source document database known for its

flexibility and horizontal scalability. In this system, Mongoose schemas simplify data validation and querying while allowing product listings across different produce categories to carry varying attributes without rigid table structures.

Implementation Detail of Modules

When the design was completed and all major considerations were addressed, the development phase of the application began. Below are the main modules of the Krishik Bazar system.

Seller Module: In this section, sellers manage their produce listings. The seller-facing controller includes handlers such as `createProduct()` for adding new listings, `updateProduct()` and `deleteProduct()` for managing existing ones, and `getSellerOrders()` for viewing incoming orders on their products.

Buyer Module: In this section, buyers browse products, manage their cart, and place orders. Key handlers include `getProducts()` for browsing and filtering the marketplace, `addToCart()` and `removeFromCart()` for cart management, and `placeOrder()` for checkout.

Authentication Module: This module ensures secure access to the system. It uses JWT-based authentication to validate users and OTP verification via nodemailer during registration. The system verifies user credentials during login and generates a secure token, stored client-side and sent as an `Authorization: Bearer` header on subsequent requests.

Order Module: This module manages the lifecycle of an order from checkout to delivery. The method `placeOrder(cartId, paymentMethod)` verifies cart contents and stock, creates order records, and clears the cart, while `updateOrderStatus()` allows tracking of the order as it moves through processing, shipped, and delivered states.

Payment Module: This module, implemented in `payment.service.js` on the frontend, currently simulates eSewa, Khalti, and Cash-on-Delivery flows in a sandbox mode, providing the interface hooks needed to wire in real payment gateway credentials for production use.

5.2 Testing

Once the source code was developed, the system was tested to identify and correct errors before deployment. The main goal of testing was to ensure that all modules function correctly and meet the system requirements.

Test Cases for Unit Testing

Unit testing focuses on verifying the smallest units of the system, such as individual modules or components. It is a white-box testing technique where each module is tested independently. In Krishik Bazar, unit testing was performed on all major modules, including authentication, product listing, cart management, and order placement. Each module was tested by providing valid and invalid inputs to check system behavior. For example, during authentication testing, correct login credentials successfully granted access, while incorrect credentials triggered error messages. Similarly, in the cart module, valid product quantities were added successfully, while out-of-stock additions were rejected.

Table 2: Testing Seller Panel

TS.No	Test Case ID	Test Description	Steps Executed	Expected Result	Actual Result	Pass/Fail
i	TC-01	Seller Login	Enter seller credentials	Redirect to dashboard	Dashboard opened successfully	Pass
ii	TC-02	Add Product	Fill product form and submit	Product created	Product created successfully	Pass
iii	TC-03	Edit Product	Update price and submit	Listing updated	Listing updated successfully	Pass
iv	TC-04	Delete Product	Click delete button	Product removed	Product removed successfully	Pass
v	TC-05	View Orders	Click orders section	Orders list displayed	Orders displayed	Pass

Table 3: Testing Buyer Panel

TS.No	Test Case ID	Test Description	Steps Executed	Expected Result	Actual Result	Pass/Fail
i	TC-01	Buyer Registration	Fill signup form	Account created	Account created successfully	Pass
ii	TC-02	OTP Verification	Enter emailed OTP	Account verified	Verification successful	Pass
iii	TC-03	Browse Products	Open marketplace page	Products displayed	Products displayed	Pass
iv	TC-04	Add to Cart	Select product and add	Item added to cart	Item added successfully	Pass
v	TC-05	Checkout	Select payment and confirm	Order placed	Order placed successfully	Pass
vi	TC-06	Track Order	Open order history page	Order status displayed	Status displayed correctly	Pass

5.3 Result Analysis

After implementation and testing, the Krishik Bazar marketplace performed successfully and met its objectives. All modules, including authentication, product listing, cart management, and order placement, worked correctly.

The system ensured secure access by validating credentials and OTPs, and prevented invalid checkout attempts such as out-of-stock or duplicate orders. Order data was recorded accurately, and overall performance was efficient with proper error handling across the API.

Thus, the system proved to be reliable, secure, and effective for connecting farmers and buyers through a direct online marketplace.

CHAPTER 6: CONCLUSION AND FUTURE RECOMMENDATIONS

6.1 Expected Outcome

The project is expected to result in a functional prototype of a farmer-to-buyer marketplace, Krishik Bazar, that enables direct produce trade over the internet.

The system will enable sellers to list produce and manage orders through a web-based interface, and will enable buyers to browse, purchase, and track their orders securely. By integrating JWT authentication and OTP verification, the system will enhance security and ensure that only genuine users can list produce or place orders.

Additionally, the system will provide a foundation for automated order management and status tracking, reducing the manual coordination typically required in traditional produce trade. The project will also demonstrate the feasibility of applying MERN stack e-commerce patterns to Nepal's agricultural sector.

6.2 Future Recommendations

While Krishik Bazar represents a functional platform, several enhancements are recommended for future iterations:

Live Payment Gateway Integration

Replacing the current sandbox payment simulation with live eSewa and Khalti API credentials would allow the platform to process real transactions securely.

Delivery and Logistics Tracking

Adding a delivery module with real-time status updates and, where feasible, courier integration would improve the end-to-end buyer experience beyond order placement.

Rating and Review System

Introducing buyer ratings and reviews for sellers and products would build trust in the marketplace and help buyers make informed purchasing decisions.

REFERENCES

1. Ministry of Agriculture and Livestock Development, Government of Nepal, "Agricultural Market Information System." Available: <https://amis.gov.np>
2. Nepal Rastra Bank, "Payment Systems and Digital Wallets in Nepal." Available: <https://www.nrb.org.np>
3. eSewa, "eSewa Merchant Integration Documentation." Available: <https://developer.esewa.com.np>
4. Khalti, "Khalti Payment Gateway API Documentation." Available: <https://docs.khalti.com>
5. MongoDB Inc., "MongoDB and Mongoose Documentation." Available: <https://www.mongodb.com/docs>
6. OpenJS Foundation, "Express.js Documentation." Available: <https://expressjs.com>